Subject-Level Calibration Heterogeneity is Stable Across LLM Families and Scales: A Statistical Audit of MMLU

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Abstract

Aggregate calibration metrics obscure a critical risk: LLMs are most overconfident in the specialized domains—legal reasoning, medical science, technical STEM—where users are least equipped to catch errors independently. We develop a statistical audit framework—a multilevel decomposition of question-level calibration gap variance, isotonic reliability curves per subject, empirical Bayes shrinkage, and Benjamini–Hochberg FDR correction—and apply it to MMLU across four instruction-tuned models from two families and three parameter scales: GPT-4o-mini, Llama-3-8B-Instruct, Qwen-2.5-1.5B-Instruct, and Qwen-2.5-7B-Instruct. Three findings emerge. (i) Subject-level ECE rankings are strongly correlated across all six model pairs (Spearman $\rho \in [0.78, 0.92]$), with cross-family agreement at matched capability stronger than within-family agreement across scale. (ii) The same subjects are robustly worst-calibrated (college-level quantitative STEM, niche factual recall, and structured ethical/legal reasoning) and robustly best-calibrated (high-school-level Social Sciences and applied/factual subjects), revealing specialization level as a meaningful axis of calibration quality. (iii) Within the Qwen family, the intercept, between-subject ICC, and entropy coefficient all change in the direction prior literature on RLHF-induced confidence saturation (Kadavath et al., 2022; Nakkiran et al., 2025) would predict, consistent with capable instruction-tuned models maintaining near-saturated confidence even when the option distribution flattens. The stability of audit findings across models supplies the structural justification for subject-aware recalibration as in Luo et al. (2025), and the audit itself supplies a transferable diagnostic for new models.

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1. Introduction

When users consult an LLM for legal guidance, medical information, or technical scientific questions, they typically lack the domain expertise to evaluate the answer independently—placing full weight on the model's expressed confidence as a reliability signal. Systematic overconfidence in these domains therefore creates a predictable, population-scale harm: users acting on high-confidence wrong answers have no reliable way to detect the error. Our findings make this concrete: across four models from two families, the most severely miscalibrated subjects are stably the professional and technical domains—`professional_law`, `virology`, `college_chemistry`—precisely where reliable confidence signals matter most. Yet the standard calibration literature reports only aggregate scalars—ECE (Guo et al., 2017), NLL, or related—that are silent about where miscalibration concentrates. We ask three questions:

1. **Where is calibration good and bad within a benchmark?**
2. **Are these patterns stable across models?**
3. **What mechanisms produce the observed heterogeneity?**

Our contribution is a statistical audit framework—multilevel decomposition, FDR correction, isotonic curves with empirical Bayes shrinkage—that produces direct answers to these questions, and an application of the framework to four instruction-tuned LLMs on MMLU (Hendrycks et al., 2021). The framework is diagnostic rather than corrective: Luo et al. (2025) demonstrates that subject-aware temperature scaling improves on global recalibration; we ask whether the structural assumption underlying that scoping decision (that miscalibration patterns are stable enough to be worth scoping over) is empirically justified.

Prior work has studied LLM calibration mostly at the aggregate level. Kadavath et al. (2022) showed that token-level logprobs carry a meaningful calibration signal on multiple-choice tasks and that instruction tuning tends to degrade it; Xiong et al. (2024) noted calibration degrades on tasks requiring professional knowledge but did not investigate subject-level structure formally. Nakkiran et al. (2025) and Yaldiz et al. (2026) document RLHF-induced confidence saturation at the aggregate level. We bring a formal statistical audit to the within-benchmark heterogeneity these works describe only in passing.

2. Method

Confidence elicitation. For each MMLU question we extract token-level log probabilities over $\{A, B, C, D\}$, softmax-renormalize, and define c_i as the maximum renormalized probability. The signed calibration gap $\text{gap}_i = c_i - y_i$ (where $y_i \in \{0, 1\}$ is correctness) is our primary outcome. GPT-4o-mini is queried via the OpenAI Batch API with `top_logprobs=20`; the open-weight models are run with 4-bit quantization on a Colab T4 GPU, with the assistant turn pre-filled by "The answer is" to score the next-token position. Test-set accuracies: 0.749, 0.606, 0.536, 0.685 for GPT, Llama, Qwen-1.5B, Qwen-7B respectively.

Multilevel decomposition. We fit a three-level mixed-effects model with questions nested in subjects nested in domains:

$$\text{gap}_{ijk} = \mu + \mathbf{x}_{ijk}^{\top}\boldsymbol{\beta} + u_k + v_{jk} + \varepsilon_{ijk}, \qquad (1)$$

where u_k is a domain random intercept, v_{jk} a subject-within-domain intercept, and $\mathbf{x}_{ijk}$ standardized question-level covariates: word count, max option length, negation indicator, and Shannon entropy of the A/B/C/D distribution ($H_i = -\sum_k p_{ik} \log p_{ik}$ in nats). Variance components are estimated by REML; the ICC decomposition reports the share of total variance at each level.

Calibration curves and ECE. For each subject we fit isotonic regression (Zadrozny & Elkan, 2002) and an unconstrained Gaussian kernel smoother ($\sigma = 1.5$ bins) on (c_i, y_i), and compute ECE as the 20-bin weighted absolute deviation between the curve and the diagonal.

Multiple testing. For each of 57 subjects we test H_0: mean calibration gap $= 0$ via a one-sample t-test on question-level gaps, then apply Benjamini–Hochberg FDR correction at $\alpha = 0.05$.

3. Results

3.1. Variance decomposition (RQ1)

Table 1 reports the ICC decomposition. Across all four models, the bulk of calibration-gap variance lies at the question level (92.2–98.1%) and domain-level variance collapses to zero. Between-subject ICCs differ by nearly a factor of four—from 1.9% (Qwen-1.5B) to 7.9% (GPT-4o-mini), with Qwen-7B intermediate at 4.9%. Within the Qwen family, ICC roughly triples as scale grows from 1.5B to 7B.

Table 1. Variance decomposition of the calibration gap. Domain-level variance is estimated at zero for every model; $\text{ICC}_{\text{subject}}$ is the share of total variance at the subject level.

Model	$\sigma^2_{\text{subject}}$	σ^2_{ε}	$\text{ICC}_{\text{subject}}$
GPT-4o-mini	0.013	0.150	0.079
Llama-3-8B-Instruct	0.004	0.190	0.020
Qwen-2.5-1.5B-Instruct	0.004	0.203	0.019
Qwen-2.5-7B-Instruct	0.009	0.177	0.049

3.2. Where calibration succeeds vs. fails (RQ1)

The same subject clusters appear at both extremes of the ECE distribution across all four models. Four subjects are top-10 most miscalibrated for every model: `virology`, `college_chemistry`, `college_physics`, `professional_law`. Top-15 expansion adds `college_mathematics`, `econometrics`, `high_school_physics`, `machine_learning`, and `moral_scenarios`. These cluster into three semantic groups: college-level quantitative STEM, niche factual recall, and structured ethical/legal reasoning.

Symmetrically, two subjects are bottom-5 best-calibrated for every model (`high_school_government_and_politics`, `high_school_psychology`); bottom-10 adds `high_school_us_history`, `marketing`, and `miscellaneous`. Two patterns:

- **Specialization level matters.** MMLU's high-school-level subjects systematically show smaller calibration gaps than their college-level counterparts (compare `high_school_psychology` to professional and college subjects in similar areas).
- **Domain category matters.** Best-calibrated subjects are dominated by Social Sciences and the "Other" (applied/factual) MMLU category; no STEM subject appears in the bottom-10 intersection across models.

Figure 1 contrasts top-5 most-miscalibrated and bottom-5 best-calibrated reliability curves for GPT-4o-mini. Curves are masked to the 1st–99th percentile of observed confidence per subject to avoid displaying isotonic extrapolation in regions with no data. The two panels reveal the heterogeneity that aggregate ECE conceals: in a single model on a single benchmark, some subjects sit on the calibration diagonal while others fall sharply below it.

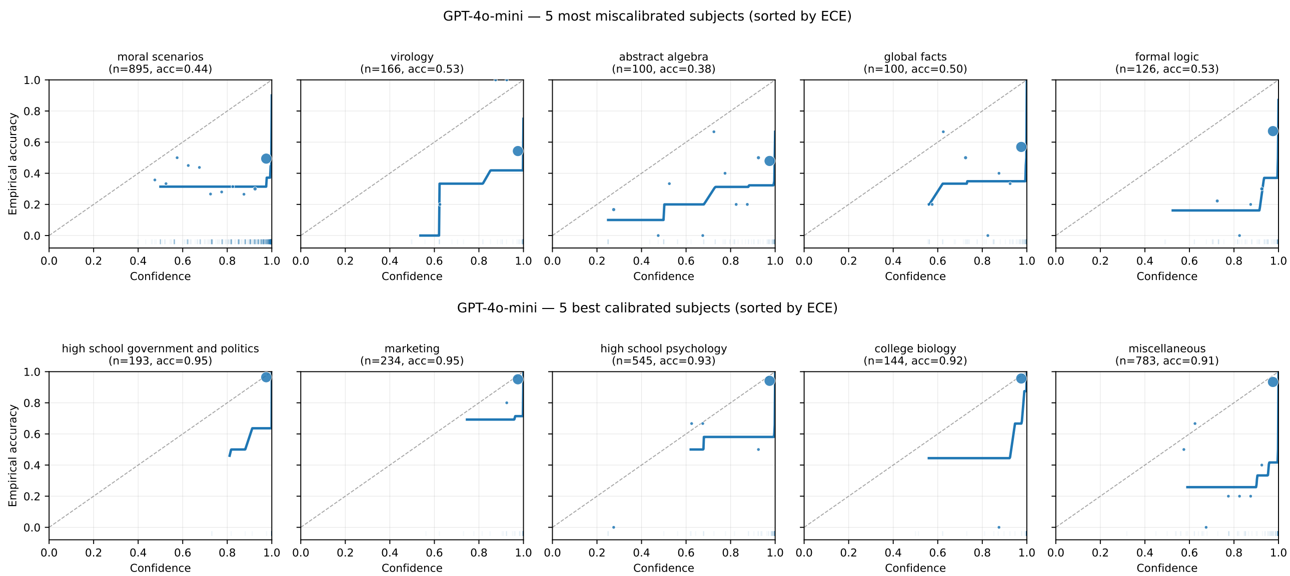

Figure 1. GPT-4o-mini reliability curves. **Top**: five most miscalibrated subjects by ECE. **Bottom**: five best-calibrated subjects. Same conventions in both: solid line is isotonic regression, plotted only on observed support; markers are 20-bin empirical accuracy sized by bin count; dashed line is the calibration diagonal.

3.3. Cross-model agreement (RQ2)

Figure 2 reports the 4×4 Spearman rank correlation matrix of subject-level ECE. All six pairwise correlations fall in $[0.78, 0.92]$. The cross-family GPT-4o-mini vs Qwen-7B pair is the highest (0.924), exceeding the within-family Qwen scaling correlation (0.849). Cross-family agreement at matched capability is therefore stronger than within-family agreement across scale—the strongest evidence in our data that subject-level miscalibration is a property of the task, not the model.

Bootstrap 95% CIs (2000 replicates): GPT vs Llama $[0.73, 0.91]$, GPT vs Qwen-1.5B $[0.64, 0.87]$, GPT vs Qwen-7B $[0.85, 0.96]$, Llama vs Qwen-1.5B $[0.73, 0.89]$, Llama vs Qwen-7B $[0.80, 0.93]$, Qwen-1.5B vs Qwen-7B $[0.76, 0.90]$.

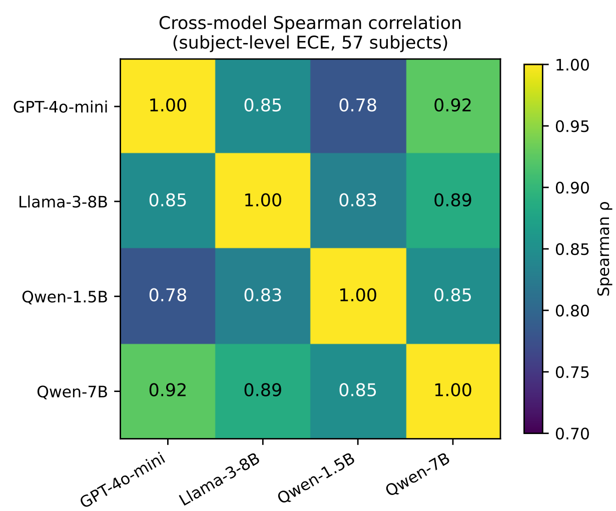

Figure 2. Spearman rank correlation matrix of subject-level ECE across the four models (57 common subjects). All pairwise correlations are between 0.78 and 0.92.

3.4. Mechanism: predictors and within-family scale (RQ3)

Table 2 reports the fixed effects from the multilevel model. Intercepts confirm large mean overconfidence (0.157–0.233). Within the Qwen family, the larger 7B model is more overconfident than the smaller 1.5B (+49% in the intercept) despite higher accuracy—consistent with Kadavath et al. (2022) on RLHF-induced confidence inflation.

Entropy: a sign flip across capability. The most informative predictor is entropy. A naïve prediction is that high option-distribution entropy should reduce the calibration gap, since top-1 confidence is mechanically lower when probability mass spreads. Yet the entropy coefficient is positive for GPT-4o-mini ($+0.016$, $p < 0.001$) and Qwen-7B ($+0.029$, $p < 0.001$), marginal for Llama, and *negative* for Qwen-1.5B (-0.010, $p = 0.018$). Figure 4 traces the mechanism by partitioning each model's questions into entropy quartiles and plotting mean confidence (solid) vs. mean accuracy (dashed). In three of four models confidence falls slower than accuracy as entropy rises, widening the gap; Qwen-1.5B is the only model where the two fall in parallel,

Table 2. Fixed effects from the three-level mixed model. Continuous covariates standardized. Significance: $^{***}p < 0.001$, $^{**}p < 0.01$, $^{*}p < 0.05$, $^{\dagger}p < 0.10$.

Predictor	GPT	Llama	Q-1.5B	Q-7B
Intercept	0.196^{***}	0.197^{***}	0.157^{***}	0.233^{***}
Word count	0.031^{***}	-0.005	0.004	0.019^{**}
Max option len.	0.001	-0.001	-0.002	-0.007
Negation	-0.001	0.007	-0.010	-0.001
Entropy	0.016^{***}	$0.008^{\dagger}$	-0.010^{*}	0.029^{***}

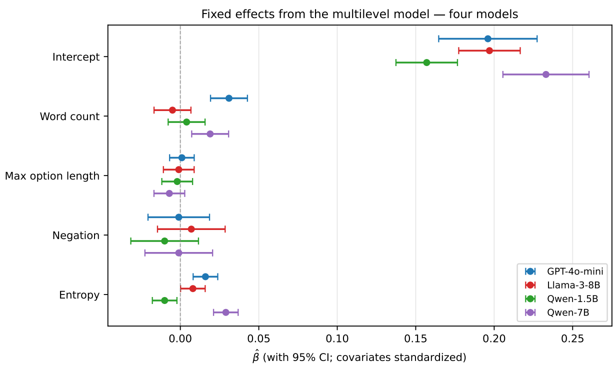

Figure 3. Fixed effect estimates with 95% confidence intervals across the four models. The entropy sign flip (negative for Qwen-1.5B, positive for the other three) is the most informative pattern.

exactly the reason its entropy coefficient is negative.

This pattern aligns with prior aggregate findings on RLHF and confidence (Kadavath et al., 2022; Nakkiran et al., 2025; Yaldiz et al., 2026)—more capable, more aggressively-aligned models maintain confidence near saturation even when the option distribution flattens. We are cautious about extending this further: the negative-coefficient finding rests on a single small checkpoint (Qwen-1.5B), and the variation is between models rather than within a controlled scaling sweep. The marginal pattern is consistent with the literature; a stronger causal claim would require additional small or base models.

3.5. FDR-corrected rankings and sensitivity

At $\alpha = 0.05$, all 57 subjects are FDR-rejected for GPT-4o-mini, Llama, and Qwen-7B; for Qwen-1.5B, 54/57 are rejected. The three subjects that fail to reject for Qwen-1.5B are `high_school_government_and_politics` ($p_{\text{adj}} = 0.37$), `sociology` ($p_{\text{adj}} = 0.24$), and `us_foreign_policy` ($p_{\text{adj}} = 0.14$)—all in Social Sciences and overlapping the robust well-calibrated set. The substantive finding is the magnitude of heterogeneity: per-model gap ranges span 6.2–13.9$\times$ from the best-calibrated to the worst-calibrated subject.

Figure 5 shows all 57 subjects sorted by mean calibration gap for GPT-4o-mini, with FDR-rejected subjects highlighted. The key takeaway is the magnitude of heterogeneity: gap ranges span 6.2–13.9$\times$ from the best-calibrated to the worst-calibrated subject across models.

Sensitivity-analysis conclusions extend uniformly: ECE rankings are stable across bin counts (Spearman ≥ 0.97 vs. 20-bin baseline), proper-scoring NLCS rankings track ECE rankings (Spearman ≥ 0.85), and two-level vs. three-level ICC estimates are numerically identical (consistent with $\sigma^2_{\text{domain}} = 0$).

4. Discussion

What the audit recovers. Aggregate ECE understates structured heterogeneity that is real (FDR-rejected, robust to bin choice, robust to proper-scoring metrics), task-driven (correlated across families and scales), and aligned with intuitive content axes (specialization level, domain category). The robust well-calibrated set (high-school-level Social Sciences and applied subjects) and the robust poorly-calibrated set (college-level quantitative STEM, niche factual recall, structured reasoning) appear in every model we examined.

Implications for recalibration. Stable cross-model targets provide the structural justification for choosing subject-aware recalibration scoping over global recalibration. We do not re-fit the corrections themselves (those would still be model-specific): Luo et al. (2025) demonstrates the operational arm directly. Our framework supplies the diagnostic upstream: when miscalibration patterns are this stable across very different training pipelines, scoping a correction at the subject level is the structurally appropriate choice. The audit findings also provide a transferable diagnostic: the robustly worst-calibrated subjects we identify can flag miscalibration on new models before deployment.

The audit as an accountability instrument. The cross-model stability of our findings has a direct governance implication: because subject-level miscalibration patterns are consistent across model families, the audit protocol is transferable without re-establishing baseline comparisons. The robustly worst-calibrated subjects—`professional_law`, `virology`, `college_chemistry`, `moral_scenarios`—constitute a natural adversarial test set for pre-deployment evaluation in high-stakes contexts. Deployers and regulators can apply this protocol as a targeted accountability checkpoint, surfacing structured overconfidence risks that aggregate ECE would not reveal and that model cards reporting only top-line metrics would obscure.

Limitations. The within-family scale comparison rests on two checkpoints (Qwen-1.5B and Qwen-7B); the entropy-coefficient sign flip rests on Qwen-1.5B alone. Replication with additional scale points (Qwen-2.5-14B) and additional

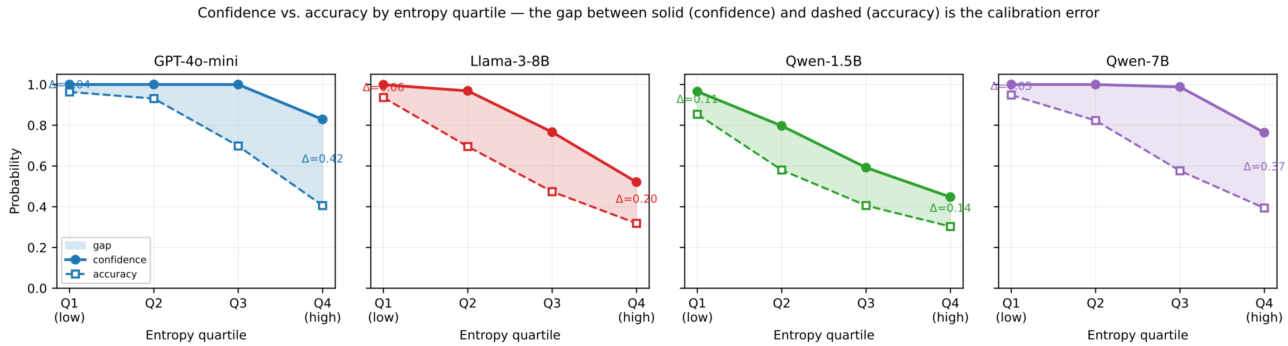

Figure 4. Confidence (solid) and accuracy (dashed) by entropy quartile per model. Shaded area is the calibration gap. Three of four models show confidence falling more slowly than accuracy as entropy rises; only Qwen-1.5B shows the two falling in parallel.

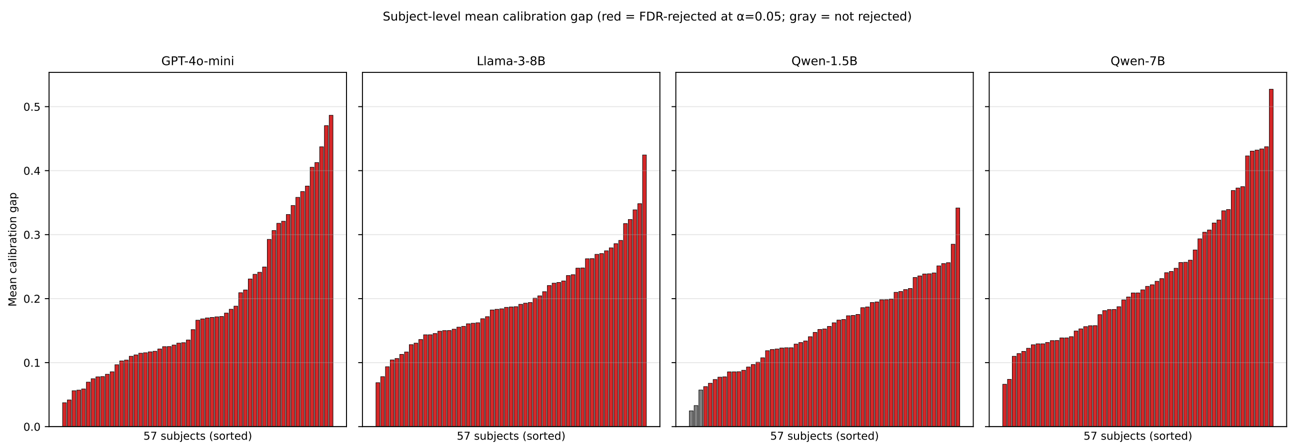

Figure 5. Mean calibration gap per subject, sorted ascending, for all four models. Red bars are FDR-rejected at $\alpha = 0.05$; gray bars are not. Three of four models reject all 57 subjects; Qwen-1.5B rejects 54 of 57.

small or base-model checkpoints (Llama-3-8B-base) would test whether the trend is monotonic. Cross-family generalization would benefit from a third family (Mistral-7B or Gemma-2-9B) at matched scale. 4-bit quantization affects three of the four models studied and may attenuate confidence; Qwen-1.5B's test-set accuracy is below its published full-precision figure, and its anomalous entropy coefficient and FDR-robustness gap may be partly artifacts of quantization. Findings are specific to MMLU's four-choice format and do not directly extend to open-ended generation.

Future work. Three extensions are most direct: (i) a continuous within-family scaling sweep, e.g., Qwen-2.5 $\{1.5\text{B}, 7\text{B}, 14\text{B}\}$, fitting intercept and entropy coefficient as functions of scale; (ii) a third model family at matched scale, strengthening the cross-family claim from two families to three; (iii) replication on harder benchmarks (MMLU-Pro) to test whether the high-school vs. college calibration gap persists at higher difficulty.

5. Conclusion

Subject-level calibration heterogeneity in LLMs is real, large, and stable. Applied/factual high-school subjects are reliably well-calibrated across all models we examined; college-level technical content, niche factual recall, and structured ethical/legal reasoning are reliably poorly calibrated. Subject-level ECE rankings are highly correlated across all six model pairs ($\rho \in [0.78, 0.92]$), including cross-family pairs at matched capability, establishing these patterns as properties of the task rather than any particular training pipeline. The audit supplies the structural justification for subject-aware recalibration scoping, a transferable pre-deployment diagnostic for new models, and a concrete accountability instrument for AI deployment in the legal, medical, and scientific domains where calibration failures carry the greatest societal cost.

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